

[Pantothenic Acid: Fact Sheet for Consumers \(original version\)](#)

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What is pantothenic acid and what does it do?

Pantothenic acid (also called vitamin B5) helps turn the food you eat into the energy you need. It's important for many functions in the body, especially making and breaking down fats.

How much pantothenic acid do I need?

The amount of pantothenic acid you need depends on your age and sex. Average daily recommended amounts are listed below in milligrams (mg).

Life Stage	Recommended Amount
Birth to 6 months	1.7 mg
Infants 7–12 months	1.8 mg
Children 1–3 years	2 mg
Children 4–8 years	3 mg
Children 9–13 years	4 mg
Teens 14–18 years	5 mg
Adults 19 years and older	5 mg
Pregnant teens and women	6 mg
Breastfeeding teens and women	7 mg

What foods provide pantothenic acid?

Pantothenic acid is naturally present in almost all foods. It is also added to some foods, including some breakfast cereals and beverages (such as energy drinks). You can get recommended amounts of pantothenic acid by eating a variety of foods, including the following:

- Beef, poultry, seafood, and organ meats
- Eggs and milk
- Vegetables such as mushrooms (especially shiitakes), avocados, potatoes, and broccoli
- Whole grains, such as whole wheat, brown rice, and oats

- Peanuts, sunflower seeds, and chickpeas

What kinds of pantothenic acid dietary supplements are available?

Pantothenic acid is available in dietary supplements containing only pantothenic acid, in B-complex dietary supplements, and in some multivitamin/mineral supplements. Pantothenic acid in dietary supplements is often in the form of calcium pantothenate or pantethine. Research has not shown that any form of pantothenic acid is better than the others.

Am I getting enough pantothenic acid?

Most people in the United States get enough pantothenic acid. However, people with a rare inherited disorder called pantothenate kinase-associated neurodegeneration can't use pantothenic acid properly. This disorder can lead to symptoms of pantothenic acid deficiency.

What happens if I don't get enough pantothenic acid?

Pantothenic acid deficiency is very rare in the United States. Severe deficiency can cause numbness and burning of the hands and feet, headache, extreme tiredness, irritability, restlessness, sleeping problems, stomach pain, heartburn, diarrhea, nausea, vomiting, and loss of appetite.

What are some effects of pantothenic acid on health?

Scientists are studying pantothenic acid to understand how it affects health. Here's one example of what this research has shown.

High cholesterol and triglyceride levels

The form of pantothenic acid called pantethine is being studied to see if it helps lower total cholesterol, low-density lipoprotein (LDL or "bad") cholesterol, and triglyceride levels. It's also being studied to see if it raises levels of high-density lipoprotein (HDL or "good") cholesterol. The results of these studies so far are promising, but more research is needed to understand the effects of pantethine dietary supplements taken alone or combined with a heart-healthy diet.

Can pantothenic acid be harmful?

Pantothenic acid is safe, even at high doses. However, taking very high doses of pantothenic acid supplements (such as 10,000 mg per day) can cause an upset stomach and diarrhea.

Does pantothenic acid interact with medications or other dietary supplements?

Pantothenic acid is not known to interact or interfere with any medicines.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if those dietary supplements might interact with your medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients.

Pantothenic acid and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about pantothenic acid?

- For general information about pantothenic acid
 - Office of Dietary Supplements (ODS) [Health Professional Fact Sheet on Pantothenic Acid](#)
- For more information on food sources of pantothenic acid
 - USDA's [FoodData Central](#)
 - Nutrient List for pantothenic acid (listed by [food](#) or by [pantothenic acid content](#)), USDA
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

calcium

A mineral found throughout the body. Calcium is needed for healthy bones and teeth, for nerves and enzymes to function properly, and for blood clotting. Calcium is found in some foods, including milk, yogurt, and cheese, and in Chinese cabbage, kale, broccoli and fortified foods, such as many drinks, tofu, and cereals.

cholesterol

A substance found throughout the body. It is made by the liver and is an important component of cells. Cholesterol is also used to make hormones, bile acid, and vitamin D. Foods that come from animals contain cholesterol, including eggs, dairy products, meat, poultry and fish. High blood levels of cholesterol increase a person's chance (risk) of developing atherosclerosis and heart disease.

deficiency

An amount that is not enough; a shortage.

diarrhea

Loose, watery stools.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder

In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

dose

The amount of medicine or other substance taken at one time or over a specific period of time.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

infant

A child younger than 12 months old.

inherit

In genetics, to receive genes that are passed from parents to their children.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

milligram

mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nausea

The uneasy feeling of having an urge to throw up (vomit).

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

pantothenate kinase-associated neurodegeneration

An inherited condition that causes harmful compounds to build up in the brain. The condition becomes noticeable in childhood or early adolescence and get worse over time. Symptoms may include muscle spasms, stiffness, trouble walking, speech problems, and vision loss. People with this condition may develop mental disabilities, personality changes, and depression.

pantothenic acid

A nutrient that is needed by the body to make energy from food and to make red blood cells, certain hormones, and the fats found in cell membranes and in tissue surrounding nerves. Pantothenic acid is found in some foods, including meat, fish, eggs, milk products, legumes, whole grains, yeast, and vegetables.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

poultry

Birds that are raised for eggs or meat, including chickens, turkeys, ducks, and geese.

prescription

A written order from a health care provider for medicine, therapy, or tests.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

triglyceride

A type of fat found in your blood. When you eat more than you need, your body turns the excess calories into triglycerides. High blood levels of triglycerides can increase your risk of heart disease, heart attack, and stroke.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition

data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

whole grain

Unprocessed seeds of edible grasses, including brown rice, buckwheat, bulgur, millet, popcorn, oats, quinoa, whole-grain barley, whole rye, whole wheat, and wild rice. Grains that are ground, cracked, or flaked can be labeled whole grain if they have the same amount of bran, germ, and endosperm (the inner part of the seed kernel) as the intact grain. Whole grains are sources of iron, magnesium, selenium, B vitamins, and dietary fiber. Eating whole grains may help lower the risk of heart disease, obesity, and type 2 diabetes.

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